LAB 13

A/B-Test Stage: a Statistical Decision before Capital Moves

Item	Detail
Topic	Topic 3: Machine Learning Models, Statistical Modelling and A/B Testing for Trading Signals
Pipeline stage	A/B test
Duration	20 min
Simulator page	http://localhost:8000/#/research/abtest?a=trend_filter&b=dip_buy
Mode	No coding. Runs ONLINE with your Alpaca paper keys (Integrations page), or OFFLINE on the demo data and the built-in paper broker. Paper trading only.

Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.

Scenario

Two portfolio managers each champion a variant for the SPY sleeve: the 200-day trend filter and the 3-day dip buy. The pipeline's rule is that capital only moves between variants when an A/B test on the same days, with the same costs, says the difference is reliable - and the decision is recorded.

Objective

Decide statistically between two variants before moving capital.

Before You Start

- Make sure the Trading Simulator is running and open http://localhost:8000 > Course Labs > Lab 13. It is installed once - see the guide 'Download, Install and Start the Trading Simulator Software' in the Learner Guide.
- Record every result you are asked to check - you will use it in the debrief and in the later labs.

Lab Steps - In the Trading Simulator

No coding is needed. Make sure the Trading Simulator is running (installed once - see the guide 'Download, Install and Start the Trading Simulator Software') and open http://localhost:8000. In the sidebar open Course Labs > Lab 13: the app shows the same steps with a Go button that opens each page pre-filled. Every step works offline on the bundled demo data and the built-in paper broker.

Step 1. Run the A/B test

Research > A/B test: A = Trend filter, B = Dip buy, SPY, alpha 0.05, 20 variants tried.

Open: http://localhost:8000/#/research/abtest?a=trend_filter&b=dip_buy (Research > A/B trend_filter vs dip_buy)

Check: Read the decision; if the CI contains 0, do not reject H0.

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Step 2. Your own variants

A = EMA crossover {"fast": 10, "slow": 30}, B = EMA crossover {"fast": 20, "slow": 50}.

Open: http://localhost:8000/#/research/abtest?a=ema_cross&b=ema_cross

Check: You can explain the Bonferroni-adjusted alpha.

Expected in the app: Welch and paired t-tests, a bootstrap Sharpe CI and a decision with a multiple-testing caution.

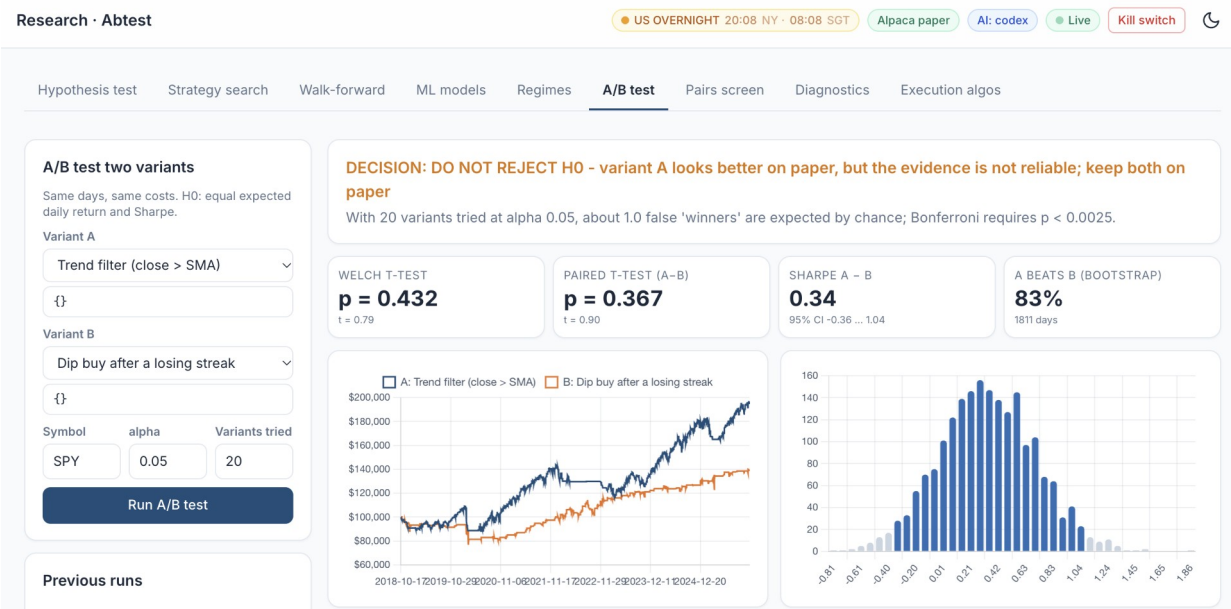


Figure: Lab 13 in the Trading Simulator.

Optional: Let AI Write the Code

This lab needs no coding. To see, run or customise the code behind it, click a link: the AI opens with the prompt filled in and writes the code for you. All prompts are also in labs/Prompt_Library.html and .pdf.

A/B test two strategy variants - run it in Google Colab
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: A/B test two strategies on SPY 2018-2025 (trade at the next day's open with a 0.05% cost per trade): A = EMA 10/30 crossover, B = EMA 20/50 crossover. Compare their daily returns with a Welch t-test, a paired t-test and a block-bootstrap 95% confidence interval of the Sharpe difference, apply a Bonferroni correction for 20 variants tried, and state the decision on H0 in one sentence.

Open in Claude | Open in ChatGPT | Open Google Colab

Debrief

A/B testing applies statistical inference to a business choice: the trend filter looks better, but the evidence cannot reject 'no difference'. Statistical tests - paired where the data are paired, bootstrapped where returns are not normal, corrected for the number of variants tried - protect capital from being moved on noise.

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